NYC TLC Operational Data Pipeline: Engineering Report

From Heterogeneous Raw Feeds to an Auditable, Dependable Ingestion & KPI Architecture

Student: Aryan Sirohi (Roll No: 10283) **Track:** Track B — NYC TLC Operations **Dataset:** Jan 2024 Yellow Taxi (2.96M trips) **Date:** September 2026

1. Problem Definition & Operational Context

The New York City Taxi & Limousine Commission (TLC) oversees yellow medallion taxi operations across the five boroughs. In January 2024 alone, over 2.96 million passenger trips were logged across separate in-vehicle vendor meters. However, fleet operations leadership lacked a consolidated, auditable view of fleet revenue realization, trip duration anomalies, and vendor reporting fidelity. Currently, raw trip records arrive partitioned by hardware vendor (Creative Mobile Technologies vs. VeriFone) in unvalidated Parquet files, taxi zone boundaries reside in a static CSV, and system metadata is exposed through an external municipal API.

The Operational Challenge: Calculating core business metrics (e.g., revenue per zone, average trip duration, speed plausibility) directly on raw vendor records results in severe distortion due to zero-passenger test runs, uncalibrated GPS spikes (>100 mph), meter clock resets, and payment dispute reversals (negative fares). This project designs, validates, and executes an automated, idempotent data engineering pipeline that ingests multi-modal sources, applies 8 strict business validation rules, logs all dropped records into an auditable rejection ledger, and outputs dependable executive KPIs without silent data loss.

2. System Architecture & End-to-End Data Flow

TLC Trip Parquet Jan 2024 (2.96M trips) HTTP Stream / CDN	**Zone Lookup CSV** 265 NYC Taxi Zones HTTP File Download	**Vendor Metadata** CMT & VeriFone Meta Local File Read	**Data Dictionary** SODA REST API JSON Metadata Endpoint

↓ **HTTP Stream & Cached Download**

INGESTION & CONTRACT ENFORCEMENT (pipeline/ingest.py)
Schema type casting, UTC timestamp normalization, MD5/size caching, missing column boundary verification

↓ **Validated Schema Ingestion**

BUSINESS VALIDATION ENGINE (pipeline/validate.py)
Evaluates 8 Deterministic Rules (Duration [1,300m], Distance [0.1,200mi], Speed [5,80mph], Passenger [1,6], Fares)

↓ **14.0% Rejections (Logged)** ↓ **86.0% Validated Trips**

AUDITABLE REJECTION LOG outputs/validation_log.csv **415,151 rows (14.0%)** Row-level failure codes (VR-01 to VR-07b)	**DIMENSIONAL ENRICHMENT & MODELING** data/processed/validated_trips.parquet **2,549,473 rows (86.0%)** — Star schema join with Zones & Vendors

↓ **Aggregation & Metric Rollup Engine**

PUBLISHED EXECUTIVE EVIDENCE REPORT (pipeline/transform.py -> outputs/metrics_report.csv)
M1: Revenue per Borough | M2: Hourly Trip Durations | M3: Vendor Share | M4: Speed Plausibility (100%) | M5: Median $/Mile

3. Source System Inventory & Retrieval Modes

To satisfy multi-modal retrieval requirements without cross-system coupling, four distinct operational sources were integrated:

Source Name	Format	Retrieval Mode	Grain	Owner / System	Identified Gaps & Operational Vulnerabilities
Yellow Trip Records	Parquet (Snappy)	HTTP download (cached)	1 row per completed trip	TLC / CloudFront CDN	Lacks referential constraints; unmetered trips present null passenger counts; negative fare entries.
Taxi Zone Lookup	CSV	HTTP download (cached)	1 row per LocationID (1-265)	NYC Open Data	LocationIDs 264 & 265 map to 'Unknown' zone; borough polygons are coarse aggregations.
Vendor Metadata	CSV (embedded)	Local file read	1 row per VendorID	TLC Technical Specs	Historical extracts contain undocumented VendorIDs (e.g. 6); mapped deterministically (1=CMT, 2=VeriFone).
TLC Data Dictionary	JSON / SODA API	REST HTTP GET	1 record per schema attribute	data.cityofnewyork.us	External network dependency; unannounced field renames can silently break downstream ETL.

4. Pipeline Dependability, Modularity & Failure Handling

The automated production pipeline is fully implemented in modular Python scripts located in pipeline/:

- **Modularity:** Strictly decoupled into ingest.py (retrieval & local disk caching), validate.py (8 named business rules), transform.py (star-schema joins & metric rollups), and config.py (system constants), orchestrated by pipeline.py.
- **Idempotency & Re-entrancy:** Running the pipeline repeatedly produces identical, deterministic results. Existing raw downloads are verified via file-size checks rather than re-downloaded, while processed artifacts are atomically overwritten rather than appended.
- **Explicit Failure Recovery:** HTTP timeouts, malformed schemas, and unreadable files immediately raise explicit exceptions with non-zero process exit codes, logging exact timestamps and stack traces to outputs/pipeline_run.log without silent failure.

NYC TLC Operational Data Pipeline: Engineering Report (Cont.)

Student: Aryan Sirohi (Roll No: 10283) **Run ID:** 20260927T133721Z **Runtime:** 43.1 Seconds **Pipeline Status:** PASSED (Exit Code: 0)

5. Business Validation Rules & Empirical Rejection Ledger

In compliance with zero silent-loss data engineering standards, every incoming record was tested against 8 deterministic rules. From 2,964,624 raw records, **2,549,473 passed (86.0%)** and **415,151 failed (14.0%)**, logged to outputs/validation_log.csv:

Rule ID	Rule Name	Condition / Boundary	Rejection Count	% of Rejects	Operational Rationale
VR-03	Speed Plausibility	Speed ∈ [5.0, 80.0] mph	239,209	57.6%	Removes extreme gridlock crawls (<5 mph) and corrupt GPS leaps (>80 mph).
VR-05	Passenger Count	Count ∈ [1, 6] (non-null)	171,687	41.4%	Excludes driver personal runs, cargo moves, and null unmetered entries.
VR-02	Trip Distance	Distance ∈ [0.1, 200.0] mi	69,770	16.8%	Eliminates zero-distance meter flips and multi-state out-of-boundary hauls.
VR-04a	Negative Fare	fare_amount < $0.00	37,448	9.0%	Dispute reversals and chargebacks must not corrupt gross trip revenue.
VR-01	Trip Duration	Duration ∈ [1.0, 300.0] min	36,935	8.9%	Discards immediate accidental cancelations (<1 min) and abandoned meters (>5 hrs).
VR-04b	Zero Fare	fare_amount == $0.00	893	0.2%	Segregates complimentary/test runs to preserve unit fare distributions.
VR-07	Timestamp Order	dropoff_time > pickup_time	870	0.2%	Rejects corrupt clock synchronizations where trip ends before it starts.
VR-06	Vendor Integrity	VendorID ∈ {1, 2}	260	0.1%	Drops uncertified hardware feeds to prevent outer join null cascades.
VR-07b	Temporal Scope	Pickup Year == 2024	15	<0.01%	Filters out uncalibrated legacy hardware emitting years 2008/2009.

** Note: Individual records can fail multiple rules simultaneously; hence rejection categories sum to >415,151 rows.*

6. FDE Judgement Call: Structured Partitioning vs. Silent Dropping

Architectural Decision: Standard tutorials routinely drop invalid rows via df[df.fare > 0]. In an enterprise production setting, this silent elimination destroys regulatory audit trails. We partitioned non-positive fares into two distinct operational classes: NEGATIVE_FARE (37,448 dispute corrections) and ZERO_FARE (893 promotional/system rides). All 415,151 rejected records were written with specific failure reason codes to outputs/validation_log.csv. This guarantees full auditability for compliance regulators while ensuring downstream KPI calculations remain strictly unbiased.

7. Epistemic Matrix: Distinguishing Facts, Assumptions & Unknowns

Category	Item & Empirical Statement	Operational Implication & System Context
FACT	2,964,624 raw records processed; 2,549,473 passed (86.0%) and 415,151 rejected (14.0%).	Verified deterministically from January 2024 Parquet download and execution logs.
FACT	Cash payment trips (payment_type=2) record exactly $0.00 tip amount.	Cash tips are received directly by drivers and are unrecorded by the meter hardware.
FACT	Vendor 2 (VeriFone) logged 1,961,857 trips (77.0%); Vendor 1 (CMT) logged 587,616 trips (23.0%).	Reflects major fleet market share concentration across technology vendors.
ASSUMPTION	Speeds <5 mph or >80 mph reflect uncalibrated GPS or stationary meter drift, not valid trips.	Ensures downstream duration and speed KPIs represent active vehicle transportation.
ASSUMPTION	January 2024 operating patterns are sufficiently representative for monthly fleet baseline.	Extreme winter weather events may suppress overall trip volume relative to summer periods.
UNKNOWN	Whether CloudFront Parquet files are static immutable replicas or batch warehouse exports.	Direct access to upstream transactional OLTP databases would permit streaming CDC ingestion.
UNKNOWN	Actual street-level routing taken between pickup and dropoff zone centroids.	Without raw GPS waypoint breadcrumbs, detour congestion cannot be separated from direct routes.

8. Key Empirical Findings & Operational Decision Support

The pipeline produced 5 verified operational metrics (saved in outputs/metrics_report.csv) supporting concrete decisions:

- **M1 (Borough Revenue Dynamics):** Manhattan accounts for 89.2% of trip volume (2,273,838 trips) generating $40.48M at an average of $17.80/trip. In contrast, Queens produces $15.39M across 247,780 trips at an average of $62.11/trip — heavily driven by flat-rate airport business trips connecting JFK and LaGuardia. Brooklyn logged 14,804 trips ($459K) and the Bronx logged 4,280 trips ($147K).
- **M2 (Hourly Congestion Patterns):** Trip volume peaks at 18:00 (183,317 trips, avg 14.4 min) and reaches its trough at 04:00 (12,489 trips). Average trip duration peaks earlier at 16:00 (17.07 min average) reflecting pre-rush gridlock before vehicles clear core business corridors.
- **M5 (Fare-per-Mile Economics):** Median fare-per-mile is highest in Manhattan ($7.28/mi) due to minimum flag drops and congestion surcharges spread over short hop distances, compared to Queens ($4.08/mi) and Brooklyn ($5.13/mi) where longer highway mileage amortizes base fees.

Executive Operational Decision: TLC leadership should deploy compliance officers to Queens airport staging facilities during the 15:00–18:00 peak revenue window to regulate flat-rate adherence, while using Manhattan duration baselines to evaluate congestion pricing impacts.